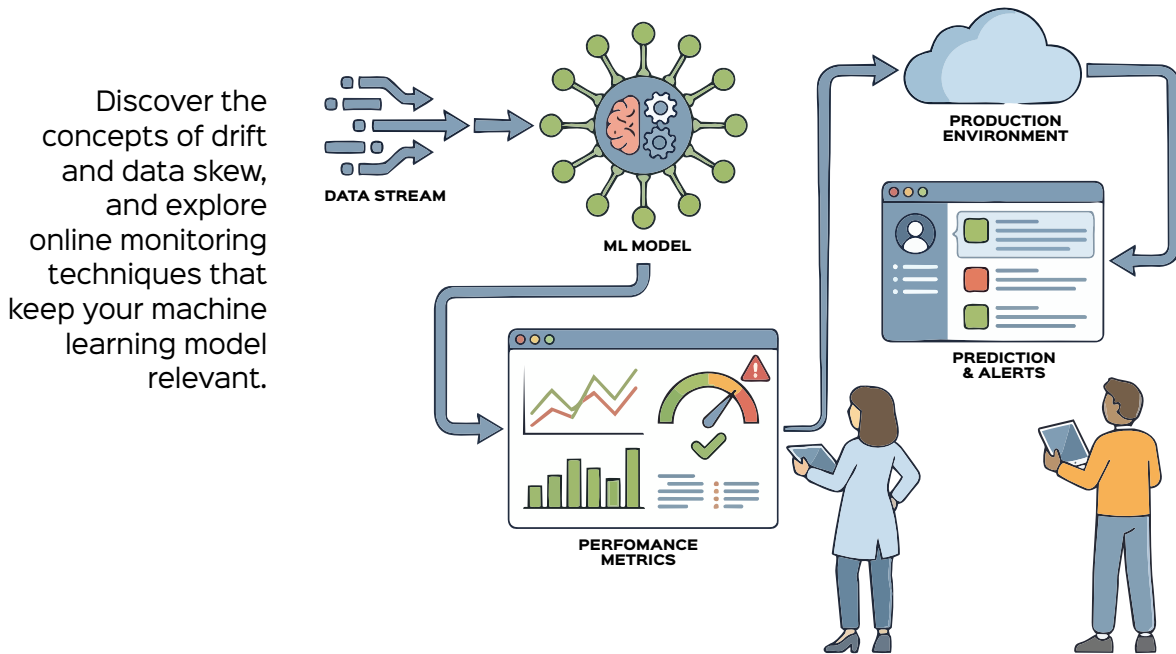


Monitoring Machine Learning in Production



In traditional software engineering, a deployment is often the end of the primary development cycle. If the unit and integration tests are passed, the software is generally expected to function until the code changes. Machine learning (ML) is fundamentally different. An ML model is a mathematical representation of a specific period. The moment it is deployed, it begins a journey towards obsolescence.

Unlike a database that crashes or an API that returns a 500 error, a decaying ML model will still return a prediction. It will still provide a 'probability score'. It looks healthy on your infrastructure dashboard, but the business value it provides is evaporating. This is why continuous monitoring is not an 'add-on' for MLOps; it is the heartbeat of the production lifecycle. Without it, you are flying blind in a changing world.

Understanding drift and skew

To monitor effectively, we must first categorise how models fail. In the world of MLOps, there are three distinct types of 'distance' between what the model expects and what it is really seeing.

Data drift (covariate shift): Data drift occurs when the statistical distribution of the input features changes over time. The model's internal logic remains 'correct' based on its training, but it is being asked to make predictions on data it has never seen before.

- **The 'new user' problem:** A fintech model trained on urban professionals is suddenly deployed in a rural market. The income levels, spending habits, and credit histories look different. The model hasn't 'failed', but it is operating outside its 'zone of competence'.

Concept drift (posterior probability shift): This is the most dangerous form of decay. Concept drift occurs when the relationship between the input and the target variable changes. Even if the input data looks exactly like the training data, the meaning of that data has shifted.

- **The post-pandemic shift:** Consider a model predicting consumer demand for travel. Before 2020, a 'low-cost flight' may have been a strong predictor of a 'guaranteed booking'. After 2020, the concept of 'safety' or 'cancellation flexibility' became more important than price. The features didn't change, but the world's behaviour did.

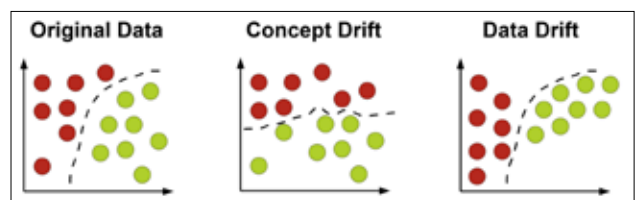


Figure 1: Data drift